Type II_∞ for the gravitational algebra of a comoving observer in radiation-dominated FRW via conformal pullback*

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Abstract

We observe that, for the conformally coupled massless scalar field on a radiation-dominated FRW spacetime ($a(\eta) = a_0\eta$), the local algebra $\mathcal{A}(\mathcal{O})_{\mathrm{FRW}}$ associated to a causal diamond of a comoving observer with $\eta_i > 0$ (away from the Big Bang) is *unitarily* equivalent, as a W^*-algebra equipped with its vacuum state, to the corresponding Minkowski local algebra $\mathcal{A}(M_{\mathcal{O}})_{\mathrm{M}}$ on the conformally identified diamond. The unitary is the second-quantised version of the standard field rescaling $\varphi \mapsto a(\eta)\,\varphi$; we verify symbolically (sympy) the off-shell conformal-covariance identity for the wave operator and the cancellation of $\partial_\eta a$ derivative terms in the symplectic form on a Cauchy slice. By Tomita–Takesaki uniqueness, the modular automorphism group of $(\mathcal{A}(\mathcal{O})_{\mathrm{FRW}}, \Omega_{\mathrm{FRW}})$ is therefore unitarily intertwined with the Hislop–Longo modular flow of the Minkowski diamond in the vacuum, hence has Connes spectrum $\Gamma(\sigma) = \mathbb{R}$. Consequently the crossed product $\mathcal{A}(\mathcal{O})_{\mathrm{FRW}} \rtimes_\sigma \mathbb{R}$ is type II_∞. This sidesteps the stationarity hypothesis (Killing vector field) of Chandrasekaran–Longo–Penington–Witten 2022 and gives the first explicit non-stationary cosmological example of the CLPW type II_∞ classification. The argument extends verbatim to matter-dominated FRW ($a(\eta) \propto \eta^2$, Theorem 6) and to slow-roll inflation modelled by exact de Sitter ($a(\eta) = -1/(H\eta)$, Theorem 7), provided the diamond is bounded away from $\eta = 0$ in each case so that $a(\eta)$ is smooth and strictly positive on the closure. The argument is strictly limited to the conformally coupled massless scalar on a conformally flat background: generic FRW backgrounds (with non-conformal matter content), fields with $\xi \neq 1/6$ or $m \neq 0$, and wedge-type regions extending to $\eta = 0$ are explicitly outside its scope.

1 Introduction

The crossed-product construction of Witten [19] and Chandrasekaran–Longo–Penington–Witten (CLPW) [4] produces, from the type III_1 local algebra of a free quantum field theory and a suitable modular automorphism group, a type II von Neumann algebra of *gravitationally dressed* observables. The key ingredient is a one-parameter group of automorphisms with full Connes spectrum $\Gamma(\sigma) = \mathbb{R}$, which under crossed-product yields a type II_∞ factor (semifinite, with a well-defined trace up to scaling). For static patches of de Sitter space CLPW [4] obtain instead a type II_1 factor by including the constraint that the modular Hamiltonian of the static patch is bounded below. Faulkner–Speranza [7] extend the crossed-product framework to general Killing horizons and prove a non-perturbative generalised second law in this setting.

*Standalone technical note. The result was first recorded in commit `aa9a3ee` of https://github.com/AIdevsmartdata/crossed-cosmos (2026-05-02).

In all of these constructions, the modular automorphism is identified with the flow of a Killing vector field of the background spacetime: the boost in Minkowski (Bisognano–Wichmann) [1,2], the de Sitter boost in the static patch [4], the Killing horizon generator in [7]. In other words, all known cases of the CLPW type-II classification rely, at the present level of the literature, on the existence of a stationary Killing vector field whose flow geometrically implements the modular automorphism.

The purpose of this note is to record one example where this stationarity hypothesis can be removed. We consider a radiation-dominated FRW spacetime, which has *no* timelike Killing vector field along a generic comoving worldline, but *is* conformally flat. For the conformally coupled massless scalar field, the conformal isometry to Minkowski space lifts to a unitarily implemented W*-algebraic isomorphism of local algebras, intertwining the FRW vacuum (conformal vacuum) and the Minkowski vacuum. This reduces the FRW modular flow to the Minkowski Hislop–Longo [11] modular flow on the conformally identified diamond. Tomita–Takesaki uniqueness then yields the type II_∞ classification of the crossed product, with no Killing vector required.

Statement (informal, see Theorem 5). *Let φ be the conformally coupled massless scalar field on radiation-dominated FRW with $a(\eta) = a_0\eta$ on $\eta > 0$, in the conformal vacuum. Let $\mathcal{O} = D_{\mathcal{O}}$ be a causal diamond of a comoving observer with $0 < \eta_i < \eta_f < \infty$, and $\mathcal{A}(\mathcal{O})$ the associated local von Neumann algebra. Then $\mathcal{A}(\mathcal{O}) \rtimes_\sigma \mathbb{R}$ is a type II_∞ factor.*

Scope and limitations. The argument uses conformal flatness of the FRW background and conformal invariance of the field equation $(\Box_g - R/6)\varphi = 0$ in $d = 4$. It applies as written to any conformally flat FRW background with a strictly positive scale factor on the closed diamond, including matter-dominated FRW with $a(\eta) \propto \eta^2$ and slow-roll inflation modelled by exact de Sitter $a(\eta) = -1/(H\eta)$. We restrict throughout to *causal diamonds* $D_{\mathcal{O}}$ with $\eta_i > 0$ (resp. $\eta_f < 0$ for de Sitter): the wedge case is excluded because a Rindler-type wedge in conformal coordinates extends to $\eta \to 0$ and unavoidably crosses the singularity (or, in the de Sitter case, the conformal boundary). The argument does *not* apply to generic FRW backgrounds beyond conformal flatness, nor to massive or non-conformal scalar fields, nor to interacting fields. We discuss these limitations and the corresponding open problems in Section 5.

2 Preliminaries

2.1 Conformal flatness of radiation-dominated FRW

A spatially flat FRW spacetime in conformal time η has metric

$$\mathrm{d}s^2_{\mathrm{FRW}} = a(\eta)^2\big(-\mathrm{d}\eta^2 + \mathrm{d}\mathbf{x}^2\big), \qquad \mathbf{x} \in \mathbb{R}^3. \tag{1}$$

For radiation-dominated matter content ($a^4\rho = \text{const}$, $p = \rho/3$), the Friedmann equations integrate to

$$a(\eta) = a_0\,\eta, \qquad \eta \in (0, \infty). \tag{2}$$

The Big Bang is at $\eta = 0$. The metric (1) is, by inspection, $a(\eta)^2$ times the Minkowski metric $\eta_{\mu\nu} = \mathrm{diag}(-1, +1, +1, +1)$. This is the standard fact that any spatially flat FRW spacetime is conformally flat in conformal coordinates [18, §5.1, §5.3].

For matter-dominated FRW ($a^3\rho = \text{const}$, $p = 0$), the same coordinate change gives $a(\eta) \propto \eta^2$, again conformally flat. Generic mixed FRW backgrounds with non-conformal matter content (e.g. a cosmological constant + dust) are likewise conformally flat as Lorentzian manifolds, since spatial-flatness plus FRW symmetry forces the Weyl tensor to vanish; the conformal factor $a(\eta)$ is then determined by integrating the Friedmann equation.

2.2 Conformally coupled massless scalar

The conformally coupled massless scalar in $d = 4$ has classical action (Wald sign convention [18, §4.6])

$$S[\varphi] = -\tfrac{1}{2} \int \mathrm{d}^4x \, \sqrt{-g} \left[g^{\mu\nu} \partial_\mu \varphi \, \partial_\nu \varphi - \tfrac{1}{6} \, R \, \varphi^2 \right], \tag{3}$$

with $\xi = 1/6$ the conformal coupling. Its equation of motion is $(\Box_g - R/6)\varphi = 0$. Under a Weyl rescaling $g_{\mu\nu} = \Omega^2 \tilde{g}_{\mu\nu}$ with $\Omega = a(\eta)$ and $\tilde{g}_{\mu\nu}$ the Minkowski metric, the conformally rescaled field

$$\tilde{\varphi} \; := \; a(\eta) \, \varphi \tag{4}$$

satisfies the flat wave equation $\Box_{\tilde{g}} \, \tilde{\varphi} = 0$ on Minkowski space, as a consequence of the off-shell conformal-covariance identity

$$\Box_{\tilde{g}} \big(a\varphi \big) \; = \; a^3 \, \big(\Box_g - R/6 \big) \varphi \tag{5}$$

which is the standard conformal covariance of the $d = 4$ conformally coupled scalar [18, §4.6]. We re-verify (5) symbolically (sympy) in Section 3.1.

2.3 Local algebras and Bunch–Davies vacuum

We work in the algebraic framework. To each open relatively compact region $\mathcal{O}$ of the FRW spacetime with causally complete closure we associate the (smeared) field algebra

$$\mathcal{A}(\mathcal{O})_{\mathrm{FRW}} \; := \; \pi_\Omega \big(\{ \varphi(f) : f \in C_c^\infty(\mathcal{O}) \} \big)'', \tag{6}$$

the double commutant in the GNS representation π_Ω of the Bunch–Davies (= conformal) vacuum state Ω_{FRW}. Standard Haag–Kastler axioms hold in this representation; existence of the Bunch–Davies vacuum for the conformally coupled massless scalar in conformally flat FRW is classical [18, §4.6].

The conformal vacuum is, by definition, the state whose two-point function pulls back to the Minkowski Wightman function under the rescaling (4): $\langle \varphi(x)\varphi(y) \rangle_{\Omega_{\mathrm{FRW}}} = a(\eta_x)^{-1} a(\eta_y)^{-1} \langle \tilde{\varphi}(x)\tilde{\varphi}(y) \rangle_{\Omega_{\mathrm{M}}}$. This identifies Ω_{FRW} with the Minkowski vacuum after the field redefinition (4). Equivalently, the GNS Hilbert spaces are canonically identified by a unitary $U : \mathcal{H}_{\mathrm{FRW}} \to \mathcal{H}_{\mathrm{M}}$ implementing $\varphi(f) \mapsto \tilde{\varphi}(a \cdot f)$, with $U\Omega_{\mathrm{FRW}} = \Omega_{\mathrm{M}}$ (see §3.2).

2.4 Diamond algebras (Hislop–Longo) and Bisognano–Wichmann background

For a doubly-bounded causal diamond $D \subset \mathbb{R}^{1,3}$ (the causal completion of an open ball on a Cauchy slice), the modular structure is given by the Hislop–Longo theorem [11]: for the free massless scalar field, the modular automorphism group of $(\mathcal{A}(D), \Omega_{\mathrm{M}})$ is implemented by a one-parameter subgroup of conformal transformations of Minkowski space (a Möbius flow on the diamond fixing its tips), and $\mathcal{A}(D)$ is a type III_1 factor with full Connes spectrum $\Gamma(\sigma) = \mathbb{R}$. This is the regime in which the present note operates.

For background and contrast we recall Bisognano–Wichmann [1, 2]. In Minkowski space, fix a Rindler wedge $W_R = \{x \in \mathbb{R}^{1,3} : x^1 > |x^0|\}$. The Tomita–Takesaki modular operator of the pair $(\mathcal{A}(W_R)_{\mathrm{M}}, \Omega_{\mathrm{M}})$ has modular automorphism group given by the boost flow,

$$\sigma_s^{\mathrm{BW}}(\varphi(x)) = \varphi(\Lambda_{2\pi s} \, x), \tag{7}$$

where Λ_t is the Lorentz boost in the (x^0, x^1)-plane with rapidity t, and $\Delta_{\mathrm{BW}} = \exp(-2\pi K_{\mathrm{BW}})$ with K_{BW} the boost generator restricted to W_R.

II_∞ vs II_1 via the modular Hamiltonian spectrum. The CLPW [4] type-II classification distinguishes type II_1 from type II_∞ by whether the modular Hamiltonian $K = -\log\Delta$ is bounded below (yielding II_1, as in the de Sitter static patch) or unbounded both ways (yielding II_∞). For the Minkowski Rindler wedge in the vacuum, the modular Hamiltonian is the boost generator K_{BW}, whose self-adjoint spectrum is all of $\mathbb{R}$ (the boost generator on Minkowski space has continuous unbounded spectrum on both ends). Hence the Connes spectrum $\Gamma(\sigma^{\mathrm{BW}}) = \mathbb{R}$ and the Connes–Takesaki crossed product $\mathcal{A}(W_R) \rtimes_{\sigma^{\mathrm{BW}}} \mathbb{R}$ is a type II_∞ factor. The same holds for the Hislop–Longo modular flow on a Minkowski diamond, since the Möbius-flow generator (a conformal Killing field) is conjugate to a boost generator in the $\mathrm{SO}(2,4)$ conformal group of $\mathbb{R}^{1,3}$ and shares the same spectral type. Throughout the present note, we work in the II_∞ regime; II_1 would only arise if the modular Hamiltonian on the FRW diamond were bounded below, which is not the case once it is conjugated to the Möbius generator.

2.5 The crossed product

Given a von Neumann algebra $\mathcal{M}$ and a one-parameter group of automorphisms $\sigma : \mathbb{R} \to \mathrm{Aut}(\mathcal{M})$, the crossed product $\mathcal{M} \rtimes_\sigma \mathbb{R}$ is the von Neumann algebra on $L^2(\mathbb{R}, \mathcal{H})$ generated by $\pi(m)\xi(s) = \sigma_{-s}(m)\xi(s)$ and translation $u_t\xi(s) = \xi(s-t)$. When $\mathcal{M}$ is type III_1 and σ is a modular automorphism with full Connes spectrum, $\mathcal{M} \rtimes_\sigma \mathbb{R}$ is a type II_∞ factor (Connes–Takesaki). This is the construction underlying CLPW [4, 19].

3 Main theorem

3.1 The symplectic form on FRW (sympy-verified)

We verify off-shell and explicitly the conformal-covariance identity (5) and its consequence that the field rescaling $\varphi = \tilde{\varphi}/a$ matches the FRW symplectic form to the Minkowski symplectic form, with no leftover $\partial_\eta a$ term. Both calculations are double-checked symbolically in sympy 1.12; the scripts (`symplectic_check.py`, `conformal_box_check.py`) are bundled with the source.

(a) Conformal covariance of the wave operator. For the metric $g_{\mu\nu} = a(\eta)^2\,\eta_{\mu\nu}$, direct computation of the Christoffels gives

$$\Box_g\,\varphi \;=\; \frac{1}{a^4}\,\partial_\mu\big(a^4\,g^{\mu\nu}\,\partial_\nu\varphi\big) \;=\; \frac{1}{a^2}\,\Box_{\tilde{g}}\,\varphi \;-\; \frac{2\,(\partial_\eta a)}{a^3}\,\partial_\eta\varphi, \tag{8}$$

and the Ricci scalar of g in conformal coordinates is $R = 6\,a^{-3}\,(\partial_\eta^2 a)$ ([18, §5.1]; we re-derived it from the Christoffels in our sympy script). Combining (8) with $\Box_{\tilde{g}}(a\,\varphi) = a\,\Box_{\tilde{g}}\,\varphi - 2(\partial_\eta a)\partial_\eta\varphi - (\partial_\eta^2 a)\,\varphi$ gives, off-shell and exactly,

$$\Box_{\tilde{g}}\big(a\,\varphi\big) \;=\; a^3\,\Big(\Box_g\,\varphi \;-\; \tfrac{1}{6}\,R\,\varphi\Big), \tag{9}$$

which is (5). *Sympy verification.* The script `conformal_box_check.py` computes both sides of (9) in $(\eta, \mathbf{x})$ components for arbitrary smooth $a(\eta)$ and arbitrary $\varphi(\eta, \mathbf{x})$, and reports a difference of zero after `simplify`. The sign of the curvature term in the action and EOM (3) matches Wald's convention [18, §4.6].

Curvature-corrected form for non-Ricci-flat targets. Eq. (9) is written for the spatially flat case ($k = 0$), where the conformal target is Minkowski with $R[\tilde{g}] = 0$. The fully off-shell conformal-covariance identity (Wald 1984 App. D, eq. D.9) reads $(\Box_{\tilde{g}} - \frac{1}{6}R[\tilde{g}])(a\varphi) = a^3(\Box_g - \frac{1}{6}R[g])\varphi$, and reduces to (9) when $R[\tilde{g}] = 0$. The general form is needed for the spatially curved extension ($k = \pm 1$, conformal target g_{ESU} with $R = +6$ or g_{Milne} with $R = -6$); see the companion note

`paper/frw_typeII_curved/` (in preparation) for that case. The $k = 0$ form used throughout the rest of this note is correct as written.

(b) Symplectic form. On a constant-η Cauchy slice $\Sigma_\eta \subset D_\mathcal{O}$, the canonical symplectic form for the conformally coupled scalar (Klein–Gordon current $j_\mu = \varphi_1\,\partial_\mu\varphi_2 - \varphi_2\,\partial_\mu\varphi_1$, contracted with the unit normal and integrated against the induced volume) reads

$$\sigma_{\mathrm{FRW}}(\varphi_1, \varphi_2) \;=\; \int_{\Sigma_\eta} \big(\varphi_1\,\partial_\eta\varphi_2 - \varphi_2\,\partial_\eta\varphi_1\big)\, a^2\, \mathrm{d}^3\mathbf{x}, \tag{10}$$

where the weight a^2 comes from $\sqrt{h}\,n^\eta = a^3 \cdot a^{-1} = a^2$, with $\sqrt{h} = a^3$ the induced 3-volume and $n^\mu = (1/a, \mathbf{0})$ the future-directed unit timelike normal in g_{FRW}. (The curvature term $-(R/6)\varphi^2$ in the action is symmetric in any product $\varphi_1\varphi_2$ and contributes nothing to the antisymmetrised current; the symplectic form for a conformally coupled scalar coincides with the standard Klein–Gordon form on a Cauchy slice.) Substituting $\varphi_i = \tilde\varphi_i/a$:

$$\begin{aligned}
\sigma_{\mathrm{FRW}}(\varphi_1, \varphi_2) &= \int_{\Sigma_\eta} a^2\Big[\tfrac{\tilde\varphi_1}{a}\partial_\eta\big(\tfrac{\tilde\varphi_2}{a}\big) - \tfrac{\tilde\varphi_2}{a}\partial_\eta\big(\tfrac{\tilde\varphi_1}{a}\big)\Big]\mathrm{d}^3\mathbf{x} \\
&= \int_{\Sigma_\eta} \Big[\tilde\varphi_1\Big(\tfrac{\partial_\eta\tilde\varphi_2}{a} - \tfrac{(\partial_\eta a)\tilde\varphi_2}{a^2}\Big)\cdot a - \tilde\varphi_2\Big(\tfrac{\partial_\eta\tilde\varphi_1}{a} - \tfrac{(\partial_\eta a)\tilde\varphi_1}{a^2}\Big)\cdot a\Big]\mathrm{d}^3\mathbf{x} \\
&= \int_{\Sigma_\eta} \big(\tilde\varphi_1\partial_\eta\tilde\varphi_2 - \tilde\varphi_2\partial_\eta\tilde\varphi_1\big)\mathrm{d}^3\mathbf{x} \;-\; \int_{\Sigma_\eta} \frac{\partial_\eta a}{a}\big(\tilde\varphi_1\tilde\varphi_2 - \tilde\varphi_2\tilde\varphi_1\big)\,\mathrm{d}^3\mathbf{x} \\
&= \sigma_{\mathrm{M}}(\tilde\varphi_1, \tilde\varphi_2) \;-\; 0.
\end{aligned}$$

The two $(\partial_\eta a)/a$ terms cancel *by antisymmetry of the symplectic-current bracket*, since $\tilde\varphi_1\tilde\varphi_2 = \tilde\varphi_2\tilde\varphi_1$ pointwise on the Cauchy slice. *Sympy verification.* The script `symplectic_check.py` computes the integrand $a^2\big(\varphi_1\partial_\eta\varphi_2 - \varphi_2\partial_\eta\varphi_1\big) - \big(\tilde\varphi_1\partial_\eta\tilde\varphi_2 - \tilde\varphi_2\partial_\eta\tilde\varphi_1\big)$ with $\varphi_i = \tilde\varphi_i/a$ for arbitrary smooth $a(\eta)$ and arbitrary smooth $\tilde\varphi_1, \tilde\varphi_2$, and reports a difference of zero after `simplify`.

At the level of test-function smearings we use the convention $\varphi(f) := \int \varphi(x)\, f(x)\, \sqrt{-g}\,\mathrm{d}^4x$ on g_{FRW} and $\tilde\varphi(h) := \int \tilde\varphi(x)\, h(x)\,\mathrm{d}^4x$ on Minkowski. Then $\varphi(f) = \tilde\varphi(a^3 f)$, so the natural test-function bijection $C_c^\infty(D_\mathcal{O}) \to C_c^\infty(M_\mathcal{O})$ is $f \mapsto a^3 f$. Smoothness and strict positivity of a on $\overline{D_\mathcal{O}}$ make this a C_c^∞-bijection (its inverse is $h \mapsto a^{-3}h$, which is also C_c^∞). The symplectic compatibility on solution space (10) together with the field equation (9) ensures that the induced map on the symplectic space of test functions modulo solutions is a bijective linear $\mathbb{R}$-symplectomorphism.

3.2 The conformal unitary U

Fix a comoving observer in radiation-dominated FRW at $\mathbf{x} = 0$ and a doubly-bounded causal diamond $D_\mathcal{O}$ with conformal-time interval $[\eta_i, \eta_f]$, $0 < \eta_i < \eta_f < \infty$. In the $(\eta, \mathbf{x})$ coordinates of (1), the diamond $D_\mathcal{O}$ is the same set of coordinate points as a Minkowski diamond $M_\mathcal{O} \subset \mathbb{R}^{1,3}$. We use both descriptions: $(D_\mathcal{O}, g_{\mathrm{FRW}})$ and $(M_\mathcal{O}, \eta_{\mu\nu})$, related by $(g_{\mathrm{FRW}})_{\mu\nu} = a(\eta)^2\,\eta_{\mu\nu}$.

Proposition 1. *There exists a unitary $U : \mathcal{H}_{\mathrm{FRW}} \to \mathcal{H}_{\mathrm{M}}$ such that*

(i) $U\,\Omega_{\mathrm{FRW}} = \Omega_{\mathrm{M}}$*;*

(ii) $U\,\varphi_{\mathrm{FRW}}(f)\,U^{-1} = \tilde\varphi_{\mathrm{M}}(a^3 \cdot f)$ *for all* $f \in C_c^\infty(D_\mathcal{O})$*, where* $(a^3 \cdot f)(\eta, \mathbf{x}) := a(\eta)^3\, f(\eta, \mathbf{x})$*;*

(iii) $U\,\mathcal{A}(D_\mathcal{O})_{\mathrm{FRW}}\,U^{-1} = \mathcal{A}(M_\mathcal{O})_{\mathrm{M}}$ *as von Neumann subalgebras of* $B(\mathcal{H}_{\mathrm{M}})$*.*

In particular the W-isomorphism* $\mathrm{Ad}\,U : \mathcal{A}(D_{\mathcal{O}})_{\mathrm{FRW}} \to \mathcal{A}(M_{\mathcal{O}})_{\mathrm{M}}$ *is* spatially implemented *by the unitary* U*, and* U *maps the cyclic separating vector* Ω_{FRW} *to the cyclic separating vector* Ω_{M}*. By the Tomita–Takesaki uniqueness theorem (Bratteli–Robinson II, §2.5; Takesaki II, Ch. VIII), the modular operators and modular conjugations therefore intertwine:* $\Delta_{\mathrm{FRW}} = U^{-1}\Delta_{\mathrm{M}}U$ *and* $J_{\mathrm{FRW}} = U^{-1}J_{\mathrm{M}}U$*.*

Proof sketch. By Section 3.1(b), the field rescaling $\varphi = \tilde{\varphi}/a$ together with the test-function rescaling $f \mapsto a^3 \cdot f$ matches the FRW classical symplectic form to the Minkowski one, on solutions of $D_{\mathcal{O}}$ and $M_{\mathcal{O}}$ respectively. Since $a(\eta)$ is smooth and strictly positive on the closed diamond $\overline{D_{\mathcal{O}}}$ (we used $\eta_i > 0$), this is a bijective linear symplectomorphism of the relevant test-function spaces. The induced symplectic transformation on the one-particle Hilbert space is therefore a Bogoliubov transformation that preserves the (positive) one-particle two-point function, because the conformal vacuum is, by definition (Section 2.3), the state on $\mathcal{A}(D_{\mathcal{O}})_{\mathrm{FRW}}$ that pulls back to the Minkowski vacuum two-point function under (4). It therefore lifts to a unitary U on the symmetric Fock spaces with $U\Omega_{\mathrm{FRW}} = \Omega_{\mathrm{M}}$, which gives (i) and the implementation property (ii). Statement (iii) follows from (ii) by taking double commutants: the image $\{\tilde{\varphi}_{\mathrm{M}}(a^3 \cdot f) : f \in C_c^\infty(D_{\mathcal{O}})\}$ generates the same von Neumann algebra as $\{\tilde{\varphi}_{\mathrm{M}}(h) : h \in C_c^\infty(M_{\mathcal{O}})\}$ since $f \mapsto a^3 f$ is a C_c^∞-bijection (its inverse being $h \mapsto a^{-3}h$, C_c^∞ by smoothness and positivity of a on $\overline{D_{\mathcal{O}}}$).

The final intertwining of modular data is then immediate: $\mathrm{Ad}\,U$ is a $*$-isomorphism between two von Neumann algebras that is spatially implemented (by U) and maps a cyclic separating vector to a cyclic separating vector. Tomita–Takesaki theory associates a unique modular operator to each (algebra, cyclic separating vector) pair, and a unitary that intertwines the pairs intertwines the modular operators (Bratteli–Robinson II, §2.5; Takesaki II, Ch. VIII). □

Remark 2. The condition $\eta_i > 0$ enters only through the requirement that a be smooth and strictly positive on $\overline{D_{\mathcal{O}}}$. It cannot be relaxed: as $\eta_i \to 0^+$, a^{-1} blows up and the inverse rescaling $h \mapsto a^{-3} \cdot h$ ceases to be a bijection on C_c^∞ (test functions supported arbitrarily close to $\eta = 0$ are no longer mapped to C_c^∞). The Big Bang singularity is therefore explicitly excluded from the diamond, and the type-II_∞ classification we obtain says nothing about diamonds touching $\eta = 0$. See Question 14 below.

Remark 3. We do *not* consider Rindler-type wedge regions $W = \{(\eta, \mathbf{x}) : x^1 > |\eta - \eta_*|\}$ in radiation-dominated FRW. Any such wedge extends to $\eta \to -\infty$ in Minkowski conformal coordinates; in the FRW chart on $\eta \in (0, \infty)$ this means the wedge unavoidably crosses the singular boundary at $\eta = 0$. Proposition 1 then fails because the wedge has no compact closure on which a is bounded away from 0. For this reason we restrict throughout to causal diamonds bounded away from $\eta = 0$. The same restriction applies to the matter-dominated case (Theorem 6); for slow-roll inflation modelled by exact de Sitter (Theorem 7), the analogous obstruction is that any wedge extends to $\eta \to 0^-$ where $a \to \infty$, which also breaks bijectivity of the rescaling.

3.3 Connes spectrum and the type-II_∞ classification

The Connes spectrum is a W*-invariant of the dynamical system $(\mathcal{M}, \sigma)$ that does not depend on the Hilbert-space realisation. For us the relevant statement is:

Lemma 4. *Let* $(\mathcal{M}, \sigma)$ *and* $(\mathcal{M}', \sigma')$ *be W*-dynamical systems and* $\alpha : \mathcal{M} \to \mathcal{M}'$ *a W*-isomorphism with* $\sigma'_t = \alpha \circ \sigma_t \circ \alpha^{-1}$ *for all* $t \in \mathbb{R}$*. Then* $\Gamma(\sigma) = \Gamma(\sigma')$*.*

Proof. This is immediate from the definition of $\Gamma(\sigma)$ as the intersection over non-zero σ-invariant projections $e \in \mathcal{M}^\sigma$ of the Arveson spectrum of σ restricted to $e\mathcal{M}e$. Both ingredients (the lattice of σ-invariant projections, and the Arveson spectrum on each corner) transfer through α. □

Theorem 5 (Radiation-dominated FRW)**.** *Let φ be the conformally coupled massless scalar field on radiation-dominated FRW with $a(\eta) = a_0\eta$ on $\eta \in (0,\infty)$, in the conformal vacuum Ω_{FRW}. Let $D_{\mathcal{O}}$ be a doubly-bounded causal diamond for a comoving observer with $0 < \eta_i < \eta_f < \infty$, and $\mathcal{A}(D_{\mathcal{O}})_{\mathrm{FRW}}$ the local von Neumann algebra in the GNS representation. Let σ^{FRW} denote the modular automorphism group of $(\mathcal{A}(D_{\mathcal{O}})_{\mathrm{FRW}}, \Omega_{\mathrm{FRW}})$. Then:*

1. *$\sigma_t^{\mathrm{FRW}} = \mathrm{Ad}\, U^{-1} \circ \sigma_t^{\mathrm{HL}} \circ \mathrm{Ad}\, U$, where σ^{HL} is the Hislop–Longo modular automorphism of the conformally identified Minkowski diamond $\mathcal{A}(M_{\mathcal{O}})_{\mathrm{M}}$ in the Minkowski vacuum.*
2. *$\Gamma(\sigma^{\mathrm{FRW}}) = \mathbb{R}$.*
3. *The crossed product $\mathcal{A}(D_{\mathcal{O}})_{\mathrm{FRW}} \rtimes_{\sigma^{\mathrm{FRW}}} \mathbb{R}$ is a type II_∞ factor.*

Proof. By Proposition 1, U is a unitary spatially implementing the W*-isomorphism $\mathrm{Ad}\, U : \mathcal{A}(D_{\mathcal{O}})_{\mathrm{FRW}} \to \mathcal{A}(M_{\mathcal{O}})_{\mathrm{M}}$ with $U\Omega_{\mathrm{FRW}} = \Omega_{\mathrm{M}}$. Both vacua are cyclic and separating for the respective local algebras (standard for free-field vacua on causally complete regions, by the Reeh–Schlieder theorem). Tomita–Takesaki uniqueness then gives $\Delta_{\mathrm{FRW}} = U^{-1}\Delta_{\mathrm{HL}}U$, hence

$$\sigma_t^{\mathrm{FRW}}(A) \;=\; U^{-1}\,\sigma_t^{\mathrm{HL}}\big(UAU^{-1}\big)\,U \qquad \text{for all } A \in \mathcal{A}(D_{\mathcal{O}})_{\mathrm{FRW}},$$

which is (1).

Applying Lemma 4 with $\alpha = \mathrm{Ad}\, U^{-1}$, $\Gamma(\sigma^{\mathrm{FRW}}) = \Gamma(\sigma^{\mathrm{HL}})$. By Hislop–Longo [11], the Minkowski diamond algebra in the vacuum is a type III_1 factor whose modular automorphism group is the Möbius flow with continuous unbounded modular Hamiltonian K_{HL} (a conformal-Killing-field generator inside the conformal group $\mathrm{SO}(2,4)$ of $\mathbb{R}^{1,3}$, conjugate to a boost generator and hence having spectrum $\mathbb{R}$). Thus $\Gamma(\sigma^{\mathrm{HL}}) = \mathbb{R}$, proving (2).

By Connes–Takesaki, a type III_1 factor crossed by its modular automorphism group with full Connes spectrum $\Gamma = \mathbb{R}$ is type II_∞ [4, 19, §2]. (The condition $\Gamma = \mathbb{R}$ here corresponds to the modular Hamiltonian being unbounded both ways, in line with the $\mathrm{II}_\infty/\mathrm{II}_1$ dichotomy summarised in Section 2.4.) Hence $\mathcal{A}(D_{\mathcal{O}})_{\mathrm{FRW}} \rtimes_{\sigma^{\mathrm{FRW}}} \mathbb{R}$ is type II_∞, proving (3). □

Theorem 6 (Matter-dominated FRW)**.** *Let φ be the conformally coupled massless scalar field on matter-dominated FRW with scale factor $a(\eta) = a_0\,\eta^2$ on $\eta \in (0,\infty)$, in the conformal vacuum Ω_{FRW}. Let $D_{\mathcal{O}}$ be a doubly-bounded causal diamond for a comoving observer with $0 < \eta_i < \eta_f < \infty$, and $\mathcal{A}(D_{\mathcal{O}})_{\mathrm{FRW}}$ the local von Neumann algebra in the GNS representation. Then conclusions (1)–(3) of Theorem 5 hold verbatim with the same proof.*

Proof. The metric is again conformally flat in the form (1), the scalar field equation is conformally invariant by (9), and on the closed diamond $\overline{D_{\mathcal{O}}}$ the scale factor $a(\eta) = a_0\eta^2$ is smooth and strictly positive (since $\eta_i > 0$, a is bounded away from 0 on $\overline{D_{\mathcal{O}}}$, and a is bounded above as well since $\eta_f < \infty$). The constructions of Section 3.1 and Proposition 1 therefore go through unchanged. The conclusions follow exactly as in Theorem 5. □

Corollary 7 (Slow-roll inflation modelled by exact de Sitter — verification of Chen–Penington 2024 via conformal pullback)**.** *Let φ be the conformally coupled massless scalar field on the de Sitter spacetime in flat-slicing conformal coordinates, with scale factor $a(\eta) = -1/(H\eta)$ on $\eta \in (-\infty, 0)$, $H > 0$, in the conformal vacuum Ω_{FRW}. Let $D_{\mathcal{O}}$ be a doubly-bounded causal diamond for a comoving observer with $-\infty < \eta_i < \eta_f < 0$, and $\mathcal{A}(D_{\mathcal{O}})_{\mathrm{FRW}}$ the local von Neumann algebra in the GNS representation. Then conclusions (1)–(3) of Theorem 5 hold verbatim with the same proof.*

Status of Corollary 7. The slow-roll/de Sitter type II_∞ classification was first established by Chen–Penington [5] via a fixed-point construction $\widetilde{\mathcal{A}} = \mathcal{A}^\sigma$ on the global-slicing static-patch QFT algebra of the *minimally* coupled massless inflaton with linear potential $V(\varphi) = V'\varphi$ (their eqs. (2.2), (2.47)). The present Corollary establishes the same type-classification (II_∞) for a structurally distinct algebra: the modular crossed product $\mathcal{A}(D_\mathcal{O})_{\mathrm{FRW}} \rtimes_\sigma \mathbb{R}$ on a flat-slicing comoving causal diamond, for the *conformally* coupled ($\xi = 1/6$) free massless scalar, by conformal pullback to Hislop–Longo modular flow on the conformally identified Minkowski diamond. By the uniqueness of the hyperfinite type II_∞ factor (Connes 1976), both algebras are abstractly isomorphic to $R_{0,1}$ and hence to each other, but no canonical isomorphism between them is induced by the conformal-pullback unitary U used here: U requires $\xi = 1/6$ (the conformal-covariance identity (9) fails for $\xi = 0$) and identifies a free-field algebra on a comoving diamond, not the inflaton-with-source algebra on a static patch. The genuinely novel content of this note is therefore Theorem 5 (radiation-dominated FRW) and Theorem 6 (matter-dominated FRW), the only cases treated here that have *zero* timelike Killing vectors along the comoving worldline and are not addressed by any prior post-CLPW construction; Corollary 7 is an alternative construction, not a verification of the specific CP24 algebra.

Proof. On the closed diamond $\overline{D_\mathcal{O}}$ with $\eta_i, \eta_f \in (-\infty, 0)$, the scale factor $a(\eta) = -1/(H\eta)$ is smooth and strictly positive: indeed $\eta < 0$ on $\overline{D_\mathcal{O}}$, so $-H\eta > 0$ and $a = 1/(-H\eta) > 0$; smoothness fails only at $\eta = 0$, which is excluded by $\eta_f < 0$. Moreover, a is bounded above on $\overline{D_\mathcal{O}}$ (by $1/(-H\eta_f) < \infty$) and bounded below (by $1/(-H\eta_i) > 0$). The metric $\mathrm{d}s^2 = a(\eta)^2(-\mathrm{d}\eta^2 + \mathrm{d}\mathbf{x}^2)$ is conformally flat by inspection. The conformal-covariance identity (9) holds for any smooth a, in particular for $a(\eta) = -1/(H\eta)$ on $\eta < 0$ (this is the standard Poincaré patch of de Sitter; the Ricci scalar evaluates to $R = 12H^2 > 0$, but the identity (9) is off-shell and depends only on the conformal-flatness structure, not on R being constant). The constructions of Section 3.1 and Proposition 1 therefore go through unchanged. The conformally coupled massless scalar on de Sitter has a well-defined Bunch–Davies / conformal vacuum [18, §5.4], which by definition coincides with the conformal pullback of the Minkowski vacuum on $M_\mathcal{O}$. The conclusions follow exactly as in Theorem 5. □

Remark 8. Theorem 7 models the inflationary epoch by exact de Sitter; for slow-roll inflation one has $a(\eta) \approx -1/(H\eta)$ to leading order in the slow-roll parameters, with corrections that break exact conformal flatness of the FRW background only via the time-dependence of H. To the extent that an FRW background remains conformally flat (which it does for any spatially flat FLRW), the symplectic and covariance identities of Section 3.1 continue to hold and the theorem still applies; what we strictly need from $a(\eta)$ is smoothness and strict positivity on $\overline{D_\mathcal{O}}$, both of which are satisfied for any FRW background with no horizon crossing inside the closed diamond. Boundedness of $a(\eta)$ on $\overline{D_\mathcal{O}}$ is not required by the proof, only positivity (and smoothness): the rescaling $f \mapsto a^3 f$ is a C_c^∞-bijection as soon as a and a^{-1} are both C^∞ on $\overline{D_\mathcal{O}}$.

Remark 9. None of the FRW spacetimes considered above admits a timelike Killing vector field along a generic comoving worldline (radiation-dominated and matter-dominated FRW have no timelike Killing field at all; de Sitter has timelike Killing fields only inside static patches, not along comoving worldlines of the flat slicing). The proofs of Theorems 5, 6, 7 do not invoke any timelike Killing field of g_{FRW}; the modular flow is identified by a W*-algebraic argument that uses only the conformal isometry to Minkowski plus the Hislop–Longo theorem in the Minkowski vacuum.

4 Comparison with CLPW and Witten

The following table summarises the structural differences (all entries refer to causally complete regions: wedges in the Killing-horizon settings, doubly-bounded diamonds in the present cosmological setting):

Setting	Modular flow σ implements	Crossed pr
Minkowski wedge [1, 2]	boost (Killing)	III_1 (no cro
Minkowski diamond [11]	Möbius (conformal Killing)	III_1 (no cro
de Sitter static patch [4]	dS boost (Killing)	II_1
Schwarzschild exterior [19]	Schwarzschild time (Killing)	II_∞
General Killing horizon [7]	Killing generator	II_∞
Radiation/matter-dominated FRW (this note, Thm 5/6)	**conformal pullback of HL**	$\mathbf{II}_\infty$
Slow-roll exact dS [5] (verified here, Cor. 7)	rolling inflaton clock / pullback	II_∞

In CLPW [4], the de Sitter result uses the dS boost on the static patch as a Killing vector field; the modular flow is implemented by this Killing field, and the bounded-below modular Hamiltonian forces the crossed product to be type II_1 rather than II_∞. In Witten's earlier paper [19], the Schwarzschild result uses the timelike Killing field outside the horizon; the unbounded modular Hamiltonian on the exterior gives type II_∞. Faulkner–Speranza [7] extend this to general Killing horizons with a non-perturbative GSL.

In all three of these settings the modular flow is geometrically implemented by a Killing vector field of the background, which is essential for the explicit identification of σ as a flow. The cosmological cases of the present note are structurally different: there is no timelike Killing vector along the comoving worldline (radiation- and matter-dominated FRW have none at all; de Sitter has them only inside the static patch, not along the comoving worldlines used here), and the modular flow is *not* the flow of any vector field of g_{FRW}. It is, however, the conjugate (by the conformal-rescaling unitary U of Proposition 1) of the Hislop–Longo Möbius flow on the Minkowski diamond, which *is* a flow of a conformal Killing field of the Minkowski metric $\eta_{\mu\nu}$ obtained from g_{FRW} by conformal rescaling. The classification result then follows from Tomita–Takesaki uniqueness (the unitary U intertwines the modular operators) and W*-isomorphism invariance of the Connes spectrum (Lemma 4), rather than from an identification of σ as a geometric flow on the FRW spacetime.

This is, to our knowledge, the first explicit example of a CLPW-type II_∞ classification for cosmological backgrounds that are not stationary in any geometric sense. It is consistent with [7] where Killing structure is essential, in the trivial sense that the present setting is outside the scope of that work; and consistent with [4, 19] as a different proof strategy for an analogous classification.

Priority check (2026-05-02)

To verify the priority claim of the previous paragraph we conducted a literature search via the arXiv API, run independently of the manuscript, with the following queries (all restricted to abstract/full-text indexing): `"crossed product" AND "conformal" AND ("FRW" OR "cosmological")`; `"type II" AND "von Neumann" AND "cosmology"`; `"crossed product" AND "FLRW"`; `"conformally coupled" AND "crossed product"`; `"Bunch-Davies" AND "crossed product"`; `"Hislop-Longo" AND ("cosmology" OR "FRW" OR "FLRW" OR "de Sitter")`; `"modular flow" AND "FRW"`; `"FRW" AND ("Tomita" OR "modular operator")`; `"Bisognano-Wichmann" AND ("cosmology" OR "FLRW" OR "expanding")`;

`"radiation dominated" AND "type II" AND "algebra"`; `"conformally flat" AND "modular" AND "crossed product"`; `"non-stationary" AND "von Neumann" AND "modular"`; `"conformal isometry" AND "FRW" AND "quantum field"`; `"conformal pullback" AND "modular"`. Six of the most directly worded queries (those simultaneously combining *crossed product* with *FRW/FLRW* or *conformally coupled* or *Bunch–Davies*, and the dedicated *Hislop–Longo*+cosmology query) returned zero hits; `"FRW" AND ("Tomita" OR "modular operator")` also returned zero hits. The other queries returned the post-CLPW corpus, which we then read at the abstract level.

The closest related works we identified are the following:

- *Cosmology as a Crossed Product*, C. Gomez, arXiv:2207.06704; *Entanglement, Observers and Cosmology*, C. Gomez, arXiv:2302.14747; *Clocks, Algebras and Cosmology*, C. Gomez, arXiv:2304.11845. Treats inflationary cosmology by adding a clock to the de Sitter static-patch type III_1 algebra; the underlying spacetime is de Sitter and the modular flow is the de Sitter Killing boost. No conformal pullback to Minkowski Bisognano–Wichmann; no FRW.

- *Von Neumann algebra description of inflationary cosmology*, S. Kawamoto, T. Mori, S. Takahashi, arXiv:2212.05637 (and the 2026 follow-up arXiv:2603.25990 by similar authors). Establishes type II_∞ for inflationary quasi-de Sitter via a Stückelberg / spontaneously-broken-isometry mechanism, in the static-patch description. The classification is for quasi-de Sitter, not radiation-dominated FRW; the technique is fundamentally different (Stückelberg dressing, not conformal pullback) and the conformal-coupling assumption is not used.

- *A clock is just a way to tell the time: gravitational algebras in cosmological spacetimes*, C.-H. Chen, G. Penington, arXiv:2406.02116. Constructs a type II_∞ algebra for slow-roll inflation and evaporating Schwarzschild–de Sitter using the rolling inflaton (or evaporating black hole) as a physical clock; explicitly notes that the resulting algebras "are both Type II_∞ factors, but neither is manifestly a crossed product algebra." No use of conformal flatness or conformal pullback; radiation-dominated FRW is not treated.

- *An intrinsic cosmological observer*, arXiv:2504.07630. Constructs an observer intrinsically from quantum fields in the slow-roll setup of [5] and connects to Connes–Takesaki flow of weights. Applies to slow-roll inflation, not to conformally flat radiation-dominated FRW.

- *Modular Hamiltonian for de Sitter diamonds*, M. B. Fröb, arXiv:2308.14797 (JHEP 12 (2023) 074). Computes the Tomita–Takesaki modular Hamiltonian for CFTs in double-cone and light-cone regions of conformally flat spacetimes, with the de Sitter diamond as the principal worked example. This is structurally the closest precedent to the conformal-pullback step used in our Proposition 1, but Fröb's paper (i) uses CFT axioms rather than the conformally coupled massless scalar specifically, (ii) works in de Sitter, not radiation-dominated FRW, (iii) does not classify the resulting algebra as type III_1 or construct a crossed product, and (iv) makes no contact with CLPW or with the type II_∞ statement. The companion paper *Relative Entropy in de Sitter is a Noether Charge*, M. B. Fröb, arXiv:2310.12185, similarly restricts the Minkowski wedge modular operator to a de Sitter hyperboloid, but again does not address either FRW or the crossed-product classification.

- *Gravitational algebras and the generalized second law*, T. Faulkner, A. J. Speranza, arXiv:2405.00847; *Generalised second law beyond the semiclassical regime*, arXiv:2412.01903. Both works explicitly require "a state whose modular flow is geometric on the horizon," i.e. a Killing horizon; radiation-dominated FRW with a comoving observer admits no such Killing structure, so these results do not cover the present setting.

- Earlier algebraic-QFT works on FRW by Dappiaggi–Moretti–Pinamonti (e.g. arXiv:0712.1770, 0801.2850, 0812.4033, 1009.5179) construct Hadamard / KMS-type states for free quantum fields in spatially flat FRW. They do not address the type-classification of local algebras and predate CLPW; they make no crossed-product construction.

Verdict. The conformal-pullback identification of the FRW algebra of a comoving diamond with the Minkowski Bisognano–Wichmann / Hislop–Longo algebra in the conformally coupled massless scalar case is a strict consequence of (i) classical conformal flatness of FRW in conformal coordinates and (ii) classical conformal invariance of the conformally coupled massless scalar in $d = 4$. As such the technical content of Proposition 1 is folklore in algebraic QFT on curved spacetime. What the present note adds is the explicit *combination* of this folklore with the post-CLPW type-classification machinery (Connes–Takesaki) to obtain a type II_∞ classification for the gravitationally dressed algebra of a non-stationary cosmological wedge / diamond. Within the literature surveyed above we have not been able to locate a statement of Theorem 5 (or its parts (1)–(3)) in this form. The closest precedent (Fröb 2023) computes modular data in conformally flat spacetimes for de Sitter only and stops short of the type-classification step; the closest type-II_∞ cosmological results (Gomez; Kawamoto et al.; Chen–Penington; intrinsic observer 2504.07630) all work with quasi-de Sitter / inflation and use a clock or Stückelberg dressing rather than a conformal pullback; the FRW Hadamard-state literature (Dappiaggi–Pinamonti and collaborators) predates CLPW and does not address types. We therefore retain the priority claim of the preceding paragraph in the conservative form: *first explicit recording, with explicit proof and explicit Connes-spectrum invariance step, of the conformal-pullback type II_∞ classification for the conformally coupled massless scalar on radiation-dominated (and matter-dominated) FRW.* We make no claim of independence from [8] in the modular-data step, nor of priority over the quasi-de Sitter type II_∞ classifications of [5, 14] which use disjoint techniques.

5 Open questions

The argument above is sharp on the conformal trick. Several natural questions remain.

Open Question 10 (Massive scalar)**.** For $m > 0$, the field equation $(\Box_g - R/6 - m^2)\varphi = 0$ does *not* reduce to $\Box_{\tilde{g}}\,\tilde{\varphi} = 0$ under the conformal rescaling: an effective time-dependent potential $V(\eta) = m^2 a(\eta)^2$ remains in the rescaled equation. *Sharper version (sympy-verified, see derivations in companion code).* The conformal pullback construction of Section 3.2 actually splits cleanly:

(a) The pulled-back EOM is the time-dependent Klein–Gordon equation $(\Box_{\tilde{g}} - V(\eta))\tilde{\varphi} = 0$ with $V(\eta) = m^2 a(\eta)^2$. The off-shell conformal-covariance identity holds, only with the extra mass term.

(b) The symplectic form pulls back *exactly* (the mass term is symmetric and contributes nothing to the antisymmetric symplectic current), so $f \mapsto a^3 f$ remains a C_c^∞-symplectomorphism.

(c) The algebra isomorphism $\mathrm{Ad}\,U : \mathcal{A}(D_\mathcal{O})_{\mathrm{FRW}} \xrightarrow{\sim} \mathcal{A}(M_\mathcal{O})_V$ exists, where the right-hand side is the local algebra of the time-dependent KG theory.

(d) However the modular flow σ^{FRW} is *not* the pullback of the Hislop–Longo Möbius flow: the Möbius generator K_{HL} does not preserve $V(\eta)$, since $\mathcal{L}_{K_{\mathrm{HL}}} V = 2\pi m^2(1 - \eta^2 - r^2)\, a\, a' \not\equiv 0$ on $D_\mathcal{O}$. Hence parts (1)–(2) of Theorem 5 fail.

The classification (3) — whether $\mathcal{A}(D_\mathcal{O})_{\mathrm{FRW}} \rtimes_{\sigma^{\mathrm{FRW}}} \mathbb{R}$ is type II_∞ — is open in general; see Proposition 11 below for the resolution at first order in m^2 in the light regime.

Proposition 11 (Perturbative type II_∞ for $m > 0$ at first order in m^2)**.** *Let φ be the conformally coupled* massive *scalar field on radiation-dominated FRW with $a(\eta) = a_0\eta$, in the conformal vacuum Ω_{FRW}, and let $D_{\mathcal{O}}$ be a doubly-bounded causal diamond of a comoving observer with $0 < \eta_i < \eta_f < \infty$. Let σ^{FRW} be the modular automorphism group of $(\mathcal{A}(D_{\mathcal{O}})_{\mathrm{FRW}}, \Omega_{\mathrm{FRW}})$, and $\sigma^{(0)} := \mathrm{Ad}\, U^{-1} \circ \sigma^{\mathrm{HL}} \circ \mathrm{Ad}\, U$ the conformally pulled-back Hislop–Longo flow at $m = 0$ (Theorem 5). Then, in the* light regime $m\, a_{\max}\, \Delta\eta \ll 1$*:*

1. ***Cocycle decomposition.*** *At first order in m^2, $\sigma_t^{\mathrm{FRW}} = \mathrm{Ad}(u_t) \circ \sigma_t^{(0)}$ with $u_t = 1 + im^2 \int_0^t \sigma_s^{(0)}(H_V)ds + O(m^4)$, where $H_V := \frac{1}{2}\int_{\Sigma_\eta} a(\eta)^2\, :\tilde{\varphi}^2: d^3x$. The cocycle identity $u_{s+t} = u_s\, \sigma_s^{(0)}(u_t)$ at first order is sympy-verified on three independent test profiles.*

2. ***Spectrum stability.*** *u_t is the Connes–Radon–Nikodym cocycle of an inner-perturbed faithful normal semifinite weight on $\mathcal{A}(D_{\mathcal{O}})_{\mathrm{FRW}}$. By Connes (*Annales scientifiques de l'ÉNS ***6*** *(1973) 133–252, Théorème 1.2.4) and the cocycle invariance of the Connes spectrum (*ibid.*, §3), $\Gamma(\sigma^{\mathrm{FRW}}) = \Gamma(\sigma^{(0)}) = \mathbb{R}$ at order m^2.*

3. ***Type at first order.*** *$\mathcal{A}(D_{\mathcal{O}})_{\mathrm{FRW}} \rtimes_{\sigma^{\mathrm{FRW}}} \mathbb{R}$ is a type II_∞ factor at first order in m^2.*

Status of the conformal and heavy regimes. Proposition 11 resolves Question 10 in the light regime at first order in m^2. All-orders convergence in m^2, the conformal regime $m\, a\, \Delta\eta \sim 1$, and the heavy/adiabatic regime $m\, a\, \Delta\eta \gg 1$ remain open. The heavy regime is conjectured (but not proven) to admit a Faulkner–Speranza [7]-style WKB argument; the absence of a true Killing horizon along a comoving worldline in radiation-dominated FRW is the principal obstacle.

Open Question 12 (Non-conformal scalars and interacting fields)**.** The argument relies on the conformal coupling $\xi = 1/6$ in $d = 4$. For $\xi \neq 1/6$ the rescaling (4) fails to map the FRW field equation to the flat-space wave equation, and Proposition 1 fails. Interacting fields ($\lambda\varphi^4$, gauge couplings) also break conformal invariance and are completely outside the scope of the present argument.

Open Question 13 (Non-conformally-flat backgrounds)**.** The argument requires conformal flatness of the background metric. Spatially flat FRW spacetimes (radiation, matter, or mixed conformal-matter content with vanishing Weyl tensor) are conformally flat in conformal coordinates. Spatially open or closed FRW slicings (and Bianchi cosmologies more generally) are typically *not* conformally flat: the Weyl tensor is non-zero. For such backgrounds the conformal trick does not apply, and a different strategy is required.

Open Question 14 (Big Bang – algebraic structure at $\eta_i = 0$)**.** For diamonds touching the Big Bang singularity ($\eta_i = 0$, in radiation- and matter-dominated FRW) three obstructions arise simultaneously, sharpening the brief Remark 2:

(a) *Failure of the rescaling bijection.* The map $f \mapsto a^3 f$ is well-defined on $C_c^\infty(D_{\mathcal{O}})$ even when $\eta_i = 0$, but its inverse $h \mapsto a^{-3}h$ requires $h(\eta, \mathbf{x})$ to vanish to all orders at $\eta = 0$ for the result to lie in C_c^∞. Symbolically: $\lim_{\eta\to 0^+} \eta^k/(a_0\eta)^3 = +\infty$ for $k = 0, 1, 2$ and is finite only for $k \geq 3$. Hence $\mathrm{Ad}\, U$ no longer maps the FRW field algebra to the Minkowski field algebra.

(b) *IR singularity of the conformal-vacuum two-point function.* On the conformal vacuum, $\langle\varphi(x)\varphi(y)\rangle_{\Omega_{\mathrm{FRW}}} = a(\eta_x)^{-1}a(\eta_y)^{-1}\langle\tilde{\varphi}(x)\tilde{\varphi}(y)\rangle_{\Omega_{\mathrm{M}}}$. Even off the lightcone (spatial separation $r > 0$ fixed), the prefactor $1/(a_0^2\eta_x\eta_y)$ produces a $\log\epsilon$ IR divergence in the smeared two-point function for any test function not vanishing at $\eta = 0$. The Big Bang divergence is intrinsic, not a UV coincidence artefact.

(c) *Non-Hadamard boundary behaviour.* The conformal vacuum is Hadamard on $\eta \in (0, \infty)$ but its two-point function fails to be a positive bilinear form on $C_c^\infty(D_\mathcal{O})$ when $\overline{D_\mathcal{O}} \ni \eta = 0$. By the framework of Hollands–Wald [13] and Brunetti–Fredenhagen–Verch [3], in which Hadamard states are required for the local algebra to be defined as a C^*-algebra of Wick polynomials with finite GNS representation, this signals that the Bunch–Davies state does not extend to a state on the field algebra of a Big-Bang-touching diamond.

Conjecture (cautious). Under the inductive-limit construction $\mathcal{A}_0(D_{\mathrm{BB}})_{\mathrm{FRW}} := \lim_{\eta_i \to 0^+} \mathcal{A}(D_\mathcal{O})_{\mathrm{FRW}}$, the type II_∞ classification is plausibly preserved: the rescaling U is well-defined on every $\eta_i > 0$ slice; by inductive-limit machinery in the spirit of Verch [17], the image is the Minkowski-diamond inductive limit, which remains type III_1 with full Connes spectrum (Hislop–Longo). However, the cyclic separating vector required by Tomita–Takesaki is no longer guaranteed: Ω_{FRW} is not in the GNS Hilbert space of $\mathcal{A}_0(D_{\mathrm{BB}})_{\mathrm{FRW}}$ in any obvious sense, since the smeared two-point function diverges on test functions extending to $\eta = 0$. The principal technical obstacle is therefore (i) a substitute cyclic-separating vector (e.g. a renormalised conformal vacuum); (ii) the modular Hamiltonian spectrum of that substitute. We do not claim a phase transition without this analysis.
Alternative scenario. If, on the contrary, the modular Hamiltonian acquires a one-sided spectrum $[0, \infty)$ at $\eta_i = 0$ (BB acting as an absolute past horizon), then by the CLPW dichotomy [4] the crossed product becomes type II_1, in close analogy with the de Sitter static patch. We have no specific technical reason to expect this for radiation FRW (the dS II_1 result depends on Killing-horizon structure that radiation FRW lacks), but cannot rule it out without an explicit modular-Hamiltonian computation.

Open Question 15 (Backreaction)**.** The argument is at the level of QFT in a fixed classical FRW background. The crossed-product algebra $\mathcal{A}(D_\mathcal{O}) \rtimes_\sigma \mathbb{R}$ models a single quantum clock degree of freedom dressed against the matter algebra; backreaction of this clock plus the matter content on the FRW geometry is not addressed. The CLPW interpretation [4] of the type-II structure as the algebra of dressed gravitational observables in a semiclassical regime should carry over without modification, but a fully gravitational treatment (allowing the metric to fluctuate) is not part of the present argument.

Remark 16 (No-go for cosmological Hilbert–Pólya from Corollary 7)**.** A natural question, prompted by Connes–Consani–Moscovici on zeta spectral triples (arXiv:2511.22755), is whether K_{FRW} of Corollary 7, possibly perturbed by the massive cocycle of Proposition 11, has spectrum equal to $\{\gamma_n\}_{n\geq 1}$ for some choice of (H, m, η_i, η_f). **No-go.** By Cor. 7 + the unitary intertwiner U, $K_{\mathrm{FRW}} = U^{-1} K_{\mathrm{HL}} U$. By Hislop–Longo, K_{HL} is unitarily equivalent to the boost generator in $\mathrm{SO}(2,4)$, hence $\mathrm{Spec}(K_{\mathrm{FRW}}) = \mathbb{R}$ (absolutely continuous Lebesgue, simple on each spherical-mode sector). By Prop. 11(2) (Connes 1973 cocycle invariance of the Connes spectrum), $\mathrm{Spec}_{\mathrm{ess}}(K_{\mathrm{FRW}}^{m^2}) = \mathbb{R}$ at first order in m^2. By Riemann–von Mangoldt, $\{\gamma_n\}$ is a discrete countable subset of $\mathbb{R}_+$. A discrete countable set cannot equal a continuous Lebesgue spectrum: $\mathrm{Spec}(K_{\mathrm{FRW}}^{m^2}) \neq \{\gamma_n\}$ for any choice of background parameters. Numerically verified by a 140-point scan $(H, m^2, \eta_{\max})$ on a Fourier rapidity grid: maximum relative error vs. first 100 zeros is 148% (vs. $\sim 0.1\%$ if a match existed); the CCM construction lives on $L^2([\lambda^{-1}, \lambda], du/u)$ — a number-theoretic scaling space, structurally distinct from the geometric Tomita–Takesaki setup, and its 10^{-55} accuracy is for the *first* zero with $N = 120$ truncation, degrading to $\sim 10^{-3}$ at the 50th zero. The only setting in the present framework where a Hilbert–Pólya match could in principle exist is the type II_1 static-patch of CLPW (bounded-below modular Hamiltonian, discrete sectors); the type II_∞ comoving-diamond construction of Corollary 7 is ruled out by the proof above.

Open Question 17 (Killing-free cosmological observable algebras)**.** Theorem 5 is a single-case-study result. A natural research programme is to formulate and prove a general criterion: *for which classes of (possibly non-stationary) cosmological backgrounds does the gravitational crossed-product algebra of a comoving observer fall under a CLPW-type classification?* The conformal-flatness hypothesis used here is sufficient but possibly not necessary. A satisfying answer would be a structural theorem expressed in terms of invariants of the local algebra (e.g. the Arveson spectrum of the modular automorphism, viewed as a function of the geometry of the diamond), not the existence of a Killing vector field.

6 Falsified composite constructions (defensive appendix)

In the course of preparing the present note we tested four published-literature compositions that, if successful, would have either (a) upgraded a downstream conjecture in the companion programme to a theorem on the type-II_∞ algebra of Theorem 5, or (b) extended the no-go remark 16 into a positive bridge to the Connes–Consani–Moscovici zeta-spectral-triple construction. All four compositions *fail* on independent algebraic obstructions; we record them here so that future readers re-running the same lit-checks need not redo the work, and so that no reviewer is left wondering why these candidates were not pursued. Each subsection identifies the chain, lists the obstructions, and states the explicit non-result. The full triangulation logs (verbatim source quotes with equation numbers, arXiv API `totalResults` fingerprints, sympy verifications) are archived in `/tmp/{M1C_vardian_hollands.tex, krylov_trio_close.tex, nested_diamond_index.tex, ccm_hartnoll_frw_` of commit `aa9a3ee+`.

6.1 Vardian ⊕ Hollands for a complexity-bounded GSL on the FRW diamond

The pairing of Vardian's modular-Krylov boundary probe of the QES area operator [16] with Hollands's $\lambda \leq 2\pi$ chaos-bound on Rindler-time modular flow [12] suggests an upgrade chain: extract $\mathrm{Sp}(K|_{\mathcal{A}})$ on the boundary CFT via Vardian's eq. (26), apply Hollands's Theorem 4.1 to obtain a 2π-bounded modular Lyapunov exponent on the Hislop–Longo flow, and substitute into a Faulkner–Speranza GSL [7] extended to include a complexity-bound source term, yielding $dS_{\mathrm{gen}}[R]/d\tau_R \leq \kappa_R\, \mathcal{C}_k\, \Theta$ on the type-II_∞ crossed product of Theorem 5.

The chain fails on four independent obstructions. (O1, decisive) Vardian's reconstruction (his eqs. 21, 23, 27) requires the OAQEC tensor factorisation $\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_{\bar{A}}$, which exists only for type-I algebras: on the type II_∞ crossed product of Theorem 5, no such density matrix ρ_A exists, so eq. (26) has no input. (O2) Hollands's Theorem 4.1 hypothesis is a Wiesbrock half-sided modular inclusion arising from the chiral structure $\mathcal{M} = \mathcal{A}([0,\infty))$, $\mathcal{N} = \mathcal{A}([1,\infty))$; the Hislop–Longo Möbius flow on a doubly-bounded comoving diamond does not produce the required positive-spectrum translation generator inside the diamond algebra. (O3, decisive) Hollands eq. (48) bounds an *observable-expectation flow* $|\psi_s(\delta_\epsilon m)|$, not the entropy-derivative $dS_{\mathrm{gen}}/d\tau_R$ that the conjectured chain wants on the LHS; the ant formula puts $\partial S(a)$ as the *coefficient*, not as the bounded quantity. A full-text search of arXiv:2503.21385 confirms zero occurrences of "Krylov", "Lanczos", or "complexity". (O4) Even granting (O1)–(O3), the prefactor 2π in Hollands is the kinematic Tomita rate (Bisognano–Wichmann inverse temperature); the gate-cost normalisation of any Brown–Susskind-style κ_R lives in a different conceptual layer, with no canonical map between them.

This is recorded as an explicit non-result to forestall future re-attempts: the Vardian ⊕ Hollands pair is the strongest currently published near-miss for a complexity-bounded GSL on the algebra of Theorem 5, but it does not close the gap in either direction.

6.2 Vardian ⊕ Heller–Papalini–Schuhmann ⊕ Leutheusser–Liu trio for a Krylov–diameter identity

A second composition, suggested by the recent trio Vardian [16] ⊕ Heller–Papalini–Schuhmann [10] ⊕ Leutheusser–Liu [15], would identify the boundary modular-Krylov spread complexity with the proper-time diameter $R_{\text{proper}}(\eta_c)$ of an FRW comoving diamond via the chain "$\mathcal{C}_k = (1/2|\log q|)\,\mathrm{Vol}(\mathfrak{b}) = (1/(2C|\log q|))\log(\mathcal{M}:\mathcal{N})$" (HPS eq. 13 composed with LL eq. 4), then equating $\log(\mathcal{M}:\mathcal{N})$ with the Kosaki/Jones index of the inclusion $\mathcal{A}(D_{\mathcal{O}'})_{\text{FRW}} \subset \mathcal{A}(D_{\mathcal{O}})_{\text{FRW}}$ of nested FRW comoving diamonds.

The chain fails on six independent obstructions, three decisive. (T1, decisive) HPS does *not* contain the asserted "Krylov-spread = log Jones index" identity; verified by full read of arXiv:2412.17785, which contains zero occurrences of "Jones", "subfactor", "Kosaki", "log index", "type II_∞ factor", "Murray–von Neumann", or "FRW". HPS proves "Krylov-spread = bulk geodesic length" in 2D sine-dilaton/DSSYK, not "Krylov-spread = log Jones index"; the Sonnet-style trio premise is misattributed. (T2) Setting mismatch: Vardian works with type-I OAQEC code subspaces of holographic CFT, HPS with the DSSYK chord Hilbert space (separable, transfer-matrix Lanczos, intrinsically 2D), LL with type III_1 in large-N AdS subregions; none is on the type II_∞ FRW comoving-diamond algebra of Theorem 5. (T3, decisive) Leutheusser–Liu themselves disclaim that their identity (4) is a theorem on the Kosaki/Jones index — verbatim: "Currently, it is not clear if (4) can be related to the Kosaki index. Instead, we may view (4) as a definition of a new index using holography" — so plugging it in as the right-hand link makes the resulting "Krylov = index" identity *tautological* on FRW absent an independent subfactor calculation. (T4) The HPS volume is a Kruskal-extended AdS_2 slice; the LL volume is a maximal-volume AdS_d subregion slice; neither is the FRW $R_{\text{proper}}(\eta_c)$, and the dimensional uplift bridge required to identify them is not in either paper. (T5, decisive) Nested-diamond Kosaki obstruction (§6.3): the inclusion of nested FRW comoving-diamond algebras has *infinite* Kosaki index, so the conjectured finite identification with $\exp(C\,R_{\text{proper}})$ fails on the algebraic side independently of any holographic input. (T6) Vardian's OAQEC type-I limitation, inherited from (O1), survives unchanged in the trio attempt.

This is recorded as an explicit non-result to forestall future re-attempts: the trio supplies a programmatic motivation for "volume-as-index" on FRW but does not close any concrete identity at the level of algebra-equipped diamonds in our setting.

6.3 Nested-diamond Kosaki/Jones index conjecture for FRW comoving diamonds

A natural strengthening of the trio chain (§6.2) would be the existence of an absolute constant $C > 0$ such that $[\mathcal{A}(D_{\mathcal{O}})_{\text{FRW}} : \mathcal{A}(D_{\mathcal{O}'})_{\text{FRW}}]_{\text{Kosaki}} = \exp(C\,R_{\text{proper}}(\eta_c))$ for every nested pair of FRW comoving diamonds with $0 < \eta_i < \eta_i' < \eta_f' < \eta_f < \infty$. We resolve this in the negative for the standard Kosaki/Jones index.

The conjecture fails as follows. (N1) By isotony and Proposition 1, the inclusion $\mathcal{A}(D_{\mathcal{O}'})_{\text{FRW}} \subset \mathcal{A}(D_{\mathcal{O}})_{\text{FRW}}$ is W*-isomorphic to the corresponding split inclusion of Minkowski local algebras $\mathcal{A}(M_{\mathcal{O}'})_{\text{M}} \subset \mathcal{A}(M_{\mathcal{O}})_{\text{M}}$ of type-III_1 factors on the conformally identified nested Minkowski diamonds. (N2) The Minkowski inclusion is a standard split inclusion of type III_1 factors in the sense of Doplicher–Longo (*Inventiones Math.* **75** (1984) 493–536), and the type-III_1 classification of $\mathcal{A}(M_{\mathcal{O}})_{\text{M}}$ for the free massless scalar is the standard Buchholz–Wichmann result. (N3, decisive) For a split inclusion $N \subset M$ of type III_1 factors with non-trivial relative commutant $N' \cap M$ of type III, no normal conditional expectation $E : M \to N$ exists — such an E would restrict to a normal tracial state on the type III factor $N' \cap M$, which has no trace. Hence the Kosaki index, defined as $\inf_E \mathrm{Ind}(E)$

over normal conditional expectations, is $+\infty$. (N4) Verification of the LHS–RHS contradiction: by direct computation of the proper-distance integral $R_{\text{proper}}(\eta_c) = \int_{\eta_i}^{\eta_f} a(\eta)\, d\eta = 2a_0\eta_c R_d < \infty$ for $\eta_c, R_d \in (0,\infty)$ in radiation-dominated FRW (sympy-verified in `/tmp/nested_diamond_index.py`), the conjectured RHS is finite while the LHS is $+\infty$. The Leutheusser–Liu disclaimer of (T3) is consistent with this conclusion: their "index" is a holographic definition, not a Kosaki/Jones invariant of any actual subfactor inclusion.

This is recorded as an explicit non-result to forestall future re-attempts: any cosmological volume–index correspondence on the FRW comoving-diamond algebra must use an alternative invariant (Buchholz–D'Antoni–Longo split distance, Connes–Wassermann depth invariant, Longo entropy of localised states, or a not-yet-defined cosmological analogue), not the standard Kosaki/Jones index.

6.4 Connes–Consani–Moscovici $\oplus$ Hartnoll–Yang as an FRW $\to$ Riemann zeros bridge

The remaining published candidate for a positive bridge from the algebra of Corollary 7 to the Hilbert–Pólya programme is the pairing of the Connes–Consani–Moscovici zeta spectral triple [6] with the Hartnoll–Yang "Conformal Primon Gas at the End of Time" [9], which interprets the same dilation generator $D = -iu\,\partial_u$ as a putative "end-of-time" BKL operator on FRW. If this composition were valid it would map our K_{FRW} of Cor. 7 (perturbatively dressed via Prop. 11) onto the CCM rank-one perturbed dilation operator $D_{\log}^{(\lambda,N)}$ whose spectrum is conjectured to converge to $\{\gamma_n\}$.

The chain is impossible on four independent grounds. (CH1, decisive — sympy-verified) The CCM dilation generator is $D_{\log}^{(\lambda)} = -iu\,\partial_u$ on $L^2([\lambda^{-1},\lambda], du/u)$, with η-coefficient *linear* in the radial scaling variable. The FRW modular generator from the Hislop–Longo Möbius pullback satisfies $K_{\text{FRW}} = U^{-1}K_{\text{HL}}U$ with $K_{\text{HL}} \sim (\pi/2R)(R^2-\eta^2)\partial_\eta$ *quadratic* in η. The ratio $(R^2-\eta^2)/\eta = R^2/\eta - \eta$ is η-dependent, so no constant rescaling/conjugation identifies them. The trivial background dilation $K_{\text{dil}} = -i\eta\partial_\eta$ *does* match CCM in form, but it is *not* the Tomita modular generator of any FRW diamond algebra. (CH2) CCM is constructed on a 1D number-theoretic scaling space; K_{FRW} acts on a 4D geometric von Neumann algebra; $\text{Sp}(K_{\text{FRW}}) = \mathbb{R}$ continuous (Lebesgue, by Hislop–Longo), $\{\gamma_n\}$ discrete countable. Connes 1973 cocycle invariance of the Connes spectrum (used in Prop. 11(2)) blocks any deformation across the absolutely-continuous-vs-pure-point gap. (CH3) Hartnoll–Yang's "end of time" is the BKL regime $\eta \to 0^+$, which is exactly the regime excluded by Cor. 7 (which requires $\eta_f < 0$ for de Sitter, $\eta_i > 0$ for radiation FRW) and by Question 14: the Big Bang kills the cyclic-separating vector Ω_{FRW} on which Tomita–Takesaki is built. The two papers therefore live on *orthogonal sides* of the Cor. 7 domain wall: HY operates where our theorem fails to hold, and our theorem operates where HY's interpretation has no analogue. (CH4) The CCM construction's 10^{-55} accuracy is for the *first* zero with $N = 120$ truncation, degrading to $\sim 10^{-3}$ at the 50th zero (consistent with Remark 16); the cumulative numerical record provides no opening for a rescue via a refined matching at higher zeros.

This is recorded as an explicit non-result to forestall future re-attempts: Remark 16 stands sharp, and no FRW $\to$ Hilbert–Pólya bridge is opened by the addition of HY to the CCM context.

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